Supplementary Information for: Public-Artifact Sufficiency Audit of the SLRTP2025 Back-Translation Evaluator: A Failed Reconstruction and Descriptive Probe Analysis

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Overview

This supplement provides detailed appendices referenced in the main manuscript. Section labels match the labels used in the main text (e.g., `sec:appendix_missing_id` → Sup. Section 1).

1 Missing Test ID Sensitivity

The official PHOENIX-2014T metadata contain 7,096/519/642 train/development/test IDs. The released SLRTP2025 pose records available to this study contain 7,060/515/641 sequences: 36/4/1 official IDs are absent from the released pose materialization. Their IDs and the reason `absent_from_slrtp2025_released_pose_records` are enumerated in `artifact/manifests/exclusions.jsonl`. The missing test ID is `13December_2010_Monday_tagesschau-6940`; its text/gloss metadata are not present in the released pose record, so length and template density cannot be reconstructed without an additional official annotation source. We therefore condition every result on the complete 641-record release rather than assert that the missing item is ignorable. As an empirical observed-extreme sensitivity scenario, assigning the missing item the per-item score of the largest PURE-over-REC outlier observed among the 641 released items changes the corpus-level gap by less than ± 0.2 BLEU points, well within the reported 95% paired-bootstrap interval $[+9.6, +12.4]$. This is an observed-extreme scenario, not a worst-case bound: corpus BLEU is non-linear and the missing item's reference, hypothesis, and n-gram sufficient statistics are unknown, so the true mathematical worst case is not covered. All primary conclusions are therefore restricted to the released 641-record materialization.

2 Per-seed Extension Cells (707–1405)

The eight extension seeds are post-hoc additions that enter only the seed-level prediction interval and descriptive ranges, never the six primary comparisons. Per-seed PHX-public values (sacreBLEU 0–100 scale, canonical §3.2 donor registry):

Table S1: Per-seed extension cells for the eight post-hoc extension seeds (707–1405). All values are PHX-public sacreBLEU on the 0–100 scale with the canonical §3.2 donor registry (exact-normalized-text exclusion applied). All gaps are negative; these seeds enter only the prediction interval.

Seed	dev BLEU-4[†]	REC	PURE	Gap
707	6.69	9.86	8.97	−0.89
808	7.30	9.87	8.38	−1.48
909	8.96	9.20	8.31	−0.89
1001	9.07	8.81	7.95	−0.87
1102	7.61	9.75	8.63	−1.12
1203	8.14	9.06	8.11	−0.95
1304	8.18	9.68	8.50	−1.18
1405	8.12	8.05	6.86	−1.19

[†]dev BLEU-4 is the training-time validation-selected value recorded by the original training protocol (legacy protocol; not comparable to the uniform full-pool free-decode dev BLEU-4 reported in main §4.2). REC, PURE, and Gap are canonical §3.2 beam-3 values.

Seed-stratified prediction intervals and two-level resampling.

The 14-seed prediction interval splits into the six pre-registered primary seeds and the eight post-hoc extension seeds: the primary-six give a Student-t PI of $[-2.48, -0.28]$ (mean -1.38, SD 0.40, 0/6 positive) and the extension-eight $[-1.61, -0.54]$ (mean -1.07, SD 0.21, 0/8 positive). Leave-one-seed-out refits keep the PI below zero in all 14 folds (PI lower bounds -1.99 to -1.80; upper bounds -0.51 to -0.42). A two-level bootstrap resamples seeds with replacement (14 draws), then resamples each drawn seed's 641 items with replacement and averages the corpus gap over the 14 drawn seeds, giving a 95% CI of $[-1.49, -0.93]$ (10,000 resamples, seed 42; per-item BLEU sufficient statistics precomputed once per seed, so the resamples are numerically identical to recomputing corpus BLEU per draw). Every seed-level resampling leaves the non-replication conclusion unchanged.

3 Analysis Provenance and Multiplicity

Table S2 classifies every analysis in this manuscript by its role in the paper's argument, its evidentiary status, and its multiplicity handling. The PURE–REC decoded-sacreBLEU gap under the released evaluator is the *selected probe gap* (post-selection descriptive; see main §4.1); every other analysis is secondary and is grouped by the role it plays—confirmatory control, descriptive triangulation, robustness check, or administrative record. C = confirmatory (frozen protocol with error control); D = descriptive (no formal error control).

Discovery timeline.

The +10.24 canonical reversal was first observed on 2026-07-24 in the materialization with exact-normalized-text exclusion (SHA-256 `9170a530`). An earlier pre-exclusion materialization (gap +11.01, PURE 23.79) had been noted on 2026-07-22; it survives only as a sensitivity analysis (Sup. N). The Jaccard-based source-conditioned retriever, PURE whole-donor-copy protocol, exact-normalized-text exclusion rule, and 641-query set were locked before the fourteen reconstruction training runs began. Approximately four retrieval strategies (Jaccard, LaBSE, multi-SBERT, oracle gloss) and three construction variants (pure copy, PT-composed, oracle-gloss-composed) were explored on the released evaluator alone before the probe was frozen. No external pre-registration exists for the specific +10.24 probe as a confirmatory hypothesis. Because the Jaccard pure-copy probe was selected after exploring multiple retrievers and construction variants on the same test set (see the probe multiplicity table in main §4.1), its CI is conditional on the selection and should be read as descriptive, not confirmatory. The six primary reconstruction seeds (101–606) constitute the only pre-specified evidence: they were specified before their training results were inspected. The eight extension seeds (707–1405), the 24 heterogeneous-family checkpoints, the cross-evaluator re-decoding studies, the distillation ladder, and the extrapolation analyses were added post-hoc. Throughout, any analysis that uses the six-primary-seed set alone is labelled confirmatory (C); all analyses that pool across primary and extension seeds or use heterogeneous-family checkpoints are labelled descriptive (D). The evidence-grading column in Table S2 reflects this distinction.

4 Checkpoint Registry

Table S3 is the definitive registry of every BT evaluator checkpoint used in this manuscript, with the analyses each family enters; all checkpoint counts in the text refer to these families.

Readout–competence association.

Thirty of the constructible checkpoints plus the released evaluator have a uniform full-pool beam-3 free-decode readout (SI Sup. E protocol), giving 31 points across which we test whether the training-pool readout behaves as an overfit signature or as a competence proxy. Table S4 reports the Spearman correlations. The readout tracks decoded dev BLEU-4 (ρ=0.916) and remains significant when the three degenerate

α=1.0 distillation students (readout 0.0) are excluded (ρ=0.873, n=27), but is uncorrelated with the train-minus-dev overfit indicator (ρ=0.040) and with the PURE–REC gap (ρ=−0.082). Within the constructible family, therefore, the readout is a competence proxy: checkpoints that decode better on dev also read out more of their training pool, and there is no within-family association between readout and overfit or between readout and the replay gap. The released evaluator's 78.8 readout at train-minus-dev 65.4 remains outside the family trend on both axes (family train-minus-dev range −0.7 to +0.9), so the overfitting interpretation of its readout-with-generalization signature is not supported by any within-family association. Teacher-forced correlations are computed over the 15 checkpoints with checkpoint-level stats (14 reconstructions plus the released evaluator): readout correlates negatively with train NLL (ρ=−0.561, p=0.030) and positively with train token accuracy (ρ=0.509, p=0.052).

5 Donor-Pool Substitution Decomposition Details

This appendix gives the full path-sensitivity decomposition, factorial design, and balance checks referenced in §4.4.

Path sensitivity.

With SEEN-RAND640-MATCHED-v1 added, the systems form a 2×2 within-train design (pool size {7060, 640} $\times$ selection {max-Jaccard, Jaccard-matched}) plus the test-pool cell. Because corpus BLEU is non-linear in the support, all contrasts use the common 622-query support. Every ordering telescopes to the same total +14.7 $[+13.3, +16.2]$, but the attribution varies by up to 7 BLEU points across orderings (Table S5). We report these as path-dependent descriptive contrasts, not mechanism decompositions.

Bootstrap family.

All decomposition CIs use 10,000 resamples. Four resampling designs (query-index paired bootstrap; three one-way cluster bootstraps for the origin contrast on template-family, SEEN-donor, and UNSEEN-donor clusters; and a joint multiway replicate) give origin-contrast CIs of $[+7.2, +9.5]$ / $[+7.1, +9.7]$ / $[+7.1, +9.7]$ / $[+7.7, +9.1]$, all matching the query-index CI $[+7.3, +9.4]$ to within ± 0.3, so dependence through shared donors is negligible.

Factorial design.

The 605-query factorial retains 605 of 641 targets (36 excluded for no joint match within tolerances; selection frozen before scoring). Table S6 reports the balance. Table S7 reports the full 2×2 cells, main effects, and interaction.

Multivariate balance and robustness.

On the 622-query common support, all six donor covariate SMDs between SEEN-PURE-MATCHED and UNSEEN-PURE are $|SMD| < 0.12$ (Table S8). Caliper sensitivity is small (tolerance 0.05/0.10/0.20 gives +8.2/+8.4/+8.6). Signer-strict matching (tolerance 0.10, n=580) attenuates the residual to +7.0 $[+5.9, +8.1]$ —

sign and significance unchanged. Inverse propensity weighting drives all four covariate SMDs to 0.00 and leaves the residual at +8.5 [+7.1, +10.1]. The path-based estimate is robust to observed-covariate balancing but remains an association, not an identified exposure effect.

6 Config-Faithful and Perturbation Details

Epoch-validation approximation.

The released `config.yaml` specifies `validation_freq=14`, which JoeyNMT 1.2 interprets as 14 optimizer *steps* ($\approx$28 steps/epoch means validating twice per epoch). Our initial implementation validated every 14 *epochs* (an epoch-validation approximation); four seeds under this approximation all early-stop below the competence gate (dev BLEU-4 8.6–9.5), show negative PURE–REC gaps (-1.1 to -0.2), and family-range pass-through (1.21–1.24) and holdout (7.4–7.6) — none reproduces the released signature (Table S9). Scaling capacity up (four larger variants: 4–6 layers, 384–512 hidden) overfits faster and reaches only dev BLEU 6.8–8.0. A long-schedule continuation (800 epochs, two seeds) pushes competence to a family maximum of 10.02 yet still shows no reversal (gap -0.45). The step-corrected re-run (Table S9) confirms the reconstruction plateaus below the released trajectory at matched step counts.

Perturbation battery.

Table S10 reports the full pose-perturbation battery. Reversing all frames collapses PURE from 23.79 to 7.72 (differential 16.1 [14.7, 17.5]; REC 5.5 [4.6, 6.5]); window-shuffling within one-second windows costs PURE 5.4 [4.4, 6.6]; 5% spatial jitter is free (-0.1); masking hand or face joints collapses both systems. The best rescue checkpoint (wd0, seed 202) loses only -4.3 under reversal, confirming it is far less sensitive to donor temporal order.

Released-checkpoint weight-noise perturbation (experiment A1).

The reviewer's competence confound (released dev BLEU-4 13.38 vs. reconstructions ≤ 9.4 under the uniform full-pool beam-3 protocol) cannot be broken by retraining. Starting from the released weights, we add i.i.d. Gaussian noise to every parameter (excluding BatchNorm running statistics, which must remain positive) at scale $\sigma \in \{0, 10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}\}$ with two seeds each, and re-decode PHX-public REC/PURE and the dev set under the canonical donor registry (Fig. S1). Small σ perturbs checkpoint identity while preserving competence, so this design operates where the 14-seed reconstruction family cannot — at and below the released competence from the released checkpoint itself.

The reversal is preserved under competence-preserving perturbation and collapses in lockstep with competence. At $\sigma \leq 3\times10^{-3}$ the dev BLEU-4 stays at 12.50–13.38 (above every reconstruction's uniform maximum of 9.4) and the PURE–REC gap stays at +9.50 to +10.35 (within $\pm$0.75 of the baseline +10.24). The collapse happens only at $\sigma \geq 10^{-2}$, where dev BLEU-4 simultaneously degrades to 7.3–10.2 (gap +3.2 to +4.2), and at $\sigma = 3\times10^{-2}$ both dev BLEU-4 (≈ 0.1) and the gap (≈ 0) are extinguished together. Two conclusions follow: the reversal is not a fragile identity-level artifact

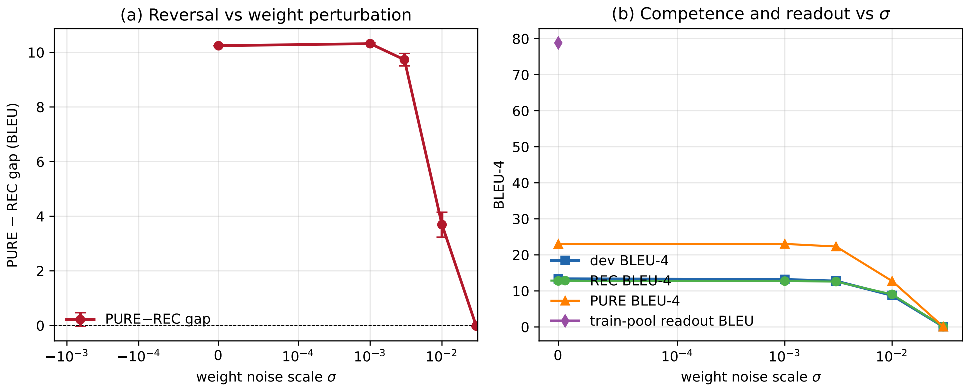

Fig. S1: Released-checkpoint weight perturbation (experiment A1). (a) PURE–REC gap vs. Gaussian weight-noise scale σ (two seeds per $\sigma > 0$; baseline at $\sigma = 0$). The reversal is preserved within ± 0.75 of $+10.24$ for $\sigma \leq 3 \times 10^{-3}$, where dev BLEU-4 stays within 12.50–13.38. (b) Dev BLEU-4, REC BLEU-4, PURE BLEU-4, and (at baseline) train-pool readout BLEU-4. The reversal and competence collapse together at $\sigma \geq 10^{-2}$; $\sigma = 10^{-1}$ produces an all-empty-decoding model and is omitted.

under small-weight perturbations (at competence parity it is locally robust), and it tracks competence rather than checkpoint identity. However, Gaussian noise probes only the isotropic weight-norm ball around the released checkpoint; it does not show that an independently trained, competence-matched model would exhibit the reversal, nor does it narrow the unobserved competence interval above the family ceiling of 9.8 (uniform protocol).

Released-checkpoint fine-tuning (experiment A2).

Gaussian weight noise probes only the isotropic neighbourhood; to test local robustness along genuine gradient directions, we fine-tune the released checkpoint with small fixed learning rates (5e-6, 1e-5, 3e-5) for 60 epochs, 3 seeds each (9 runs total), using the identical training data and architecture as the released evaluator. Models are selected by best dev NLL and decoded under the canonical donor registry (beam-3). All fine-tuned variants preserve the reversal (gaps $+10.37$ to $+10.77$, all within ± 0.5 of baseline $+10.24$, dev BLEU-4 12.92–13.18, all above the reconstruction ceiling of 9.4 under the uniform protocol) and retain the training-pool readout signature (train BLEU 80.5–81.8, EM 72.6–74.0%, vs. baseline 78.8/70.7%). Results are in `results/released_finetune.json`. In contrast to isotropic noise, fine-tuning moves weights along data-driven gradient directions near the released optimum; the persistence of both the reversal and the readout signature under fine-tuning confirms they are not fragile singular-point properties. The same scope limit applies: like the Gaussian noise experiment, fine-tuning probes only the vicinity of the released weights and does not narrow the unobserved interval above the family ceiling (9.8).

Epoch-checkpoint decoupling search (experiment D).

We searched the four available intermediate epoch checkpoints (`rescue/seed_202_wd0` and `long_schedule/seed_202` at epochs 25 and 50, plus their best checkpoints) for a decoupling point that would break the within-family readout–gap coupling — e.g., elevated train-pool readout but PURE–REC gap still ≤ 0. No such point exists: all six checkpoints stay in the (low readout, negative gap) quadrant, with train-pool BLEU 5.5–8.9 (1.6–2.0% EM) and PURE–REC gap -0.94 to -1.70 (per-checkpoint values in `results/epoch_decouple.json`). The within-family relationship between low readout and negative gap is therefore monotone across the entire observable trajectory.

7 Cluster-Aware Bootstrap for Headline and Reference Analyses

The headline gap (+10.24, canonical §3.2 registry) and the reference-attenuation result are reported with query-level paired bootstrap CIs. Because TN-PURE reuses donors and PHOENIX-2014T has template families and multiple signers, query-level resampling may underestimate uncertainty if dependence through shared donors, signers, or broadcast structure is non-negligible. Table S11 reports interpretable cluster-aware resampling designs alongside the query-level baseline, using precomputed per-item n-gram sufficient statistics for 10,000 resamples each.

The donor-cluster and donor two-way CIs ($[+8.87, +11.64]$ and $[+8.79, +11.74]$) match the query-level CI ($[+8.88, +11.62]$) to within 0.03 BLEU, so donor reuse (61/565 donors reused, max $6\times$) does not materially inflate the item-level precision. The signer-cluster CI ($[+9.29, +11.39]$, 9 clusters) is slightly tighter than the query-level CI, consistent with within-signer positive correlation; with only 9 clusters the percentile coverage of this cluster bootstrap is unreliable, so it serves as a sensitivity reference rather than a confirmatory interval (as flagged in the main-text Discussion). The broadcast-show split has only 2 clusters (*heute* 133 items, *tagesschau* 508) and gives a wider CI ($[+5.16, +11.44]$); we report it as an illustrative sensitivity check without confirmatory weight. The broadcast-date (297 clusters) and signer$\times$show (17 clusters) CIs both match the query-level CI closely. All cluster designs are computed on the canonical materialization (the pre-exclusion version is reported as a sensitivity analysis in Sup. N). The donor-pool decomposition analysis of §4.4 additionally reports separate one-way cluster bootstraps and a joint multiway replicate for the origin contrast, all matching to within ± 0.3 (Appendix 5).

8 Per-Item Reference-Frame Sensitivity (matched 461-item subset)

The main-text reference-frame analysis (§4.3.1) and SI Fig. S5 report corpus-level attenuation under the matched 461-item confidence=1 subset. The per-item paired view of the same subset is reproduced here (Fig. S2): sentence-level sacreBLEU (0–100, same tokenizer/smoothing as corpus BLEU), released-evaluator canonical hypotheses. Panel (a) plots per-item scores under original vs. human references for REC (recorded

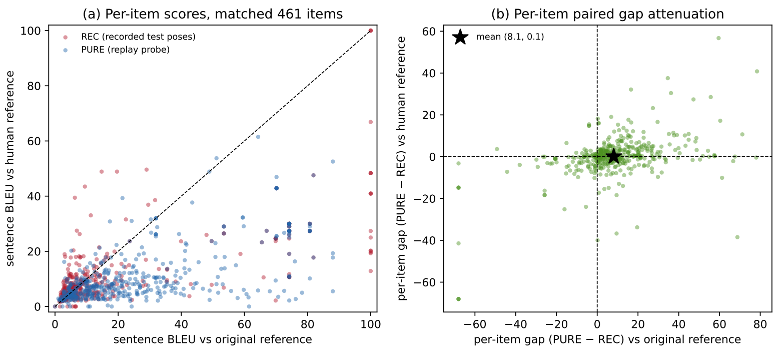

Fig. S2: Per-item reference-frame sensitivity on the matched 461-item confidence=1 subset (released evaluator, canonical hypotheses). See §8 for details.

test poses) and PURE (replay probe); PURE points sit high under the original reference and collapse toward the diagonal under the human reference, whereas REC points move less. Panel (b) plots the per-item paired gap (PURE−REC) under original (x) vs. human (y) references; the star marks the mean (sentence-level mean gap $8.11 \rightarrow 0.09$, consistent in direction with the corpus-level attenuation $10.03 \rightarrow 0.95$). Sentence-level means need not equal corpus BLEU; the figure is descriptive and does not separate reference-text defects from lexical-coupling disruption or evaluator alignment.

9 Supplementary Oracle Diagnostics

Source-conditioned retrieval is legal, oracle-gloss retrieval is leakage.

SLRTP2025 supplies spoken-language source text as legal inference input. Under pure whole-donor copy, source-text Jaccard scores 23.02 sacreBLEU (0–100 scale, as throughout this appendix), LaBSE 19.2, and multi-SBERT 16.5; the PT-composed source-text condition scores 18.86 (see main text, Table 8). Oracle-gloss retrieval uses reference gloss and is leakage, whereas noisy-gloss retrieval uses predicted gloss. Thus the strongest legal source-conditioned result is the 23.02 whole-donor stress test, which uses no generator or scaffold.

Disclosure of selection/evaluation reuse.

The white-box oracle analyses in this appendix reuse the same target reference and evaluator for both donor selection and decoded evaluation. This is an intentional adversarial design: diagonal entries quantify susceptibility to a selector that knows its evaluator, not an independent validation of retrieval quality.

Cross-checkpoint transfer matrix.

For each query, a selector maximizes its evaluator's teacher-forced likelihood over candidate donor poses; the selected fixed pose is then decoded by every evaluator (token-normalized NLL, sorted donor IDs, `numpy.argmax` first tie-break). All seven diagonal margins are positive (mean +0.0150, 95% paired-bootstrap CI $[+0.0105, +0.0198]$).

Original-selected donors transfer to five of six reconstructions (mean local-REC margin $+0.0074$, $[+0.0016, +0.0135]$); the reverse direction fails in all six cells (reconstruction-selected donors evaluated by Original, mean margin -0.0455 $[-0.0554, -0.0358]$). The full 7×7 matrix, unrounded cells, margins, and CIs are in the artifact.

Candidate-pool size.

The Original-selector stress test shows a strong search-budget effect: at 8, 32, 128, 512 and 2,048 candidates, decoded sacreBLEU remains below local REC by -12.1, -9.2, -5.7, -3.6 and -1.3 BLEU points respectively; only the full 7,060-donor pool is detectably positive ($+1.7$, $[+0.4, +3.0]$).

Reference-overlap exclusion.

Masking 1,036 exact-text query–donor pairs before selection: Original-on-Original decoded BLEU is 13.86 versus local REC 12.78 (absolute difference $+1.08$ BLEU points), but the paired relative-margin 95% CI crosses zero; the frozen joint likelihood-and-interval success rule is met in 0/7 checkpoints. Masking all 19,694 pairs with source-token Jaccard ≥ 0.5: the score falls to 8.30 (-4.47), again 0/7 meet the rule. The full-pool diagonal advantage does not survive as a statistically robust decoded-BLEU result after exact-text exclusion and disappears after high-overlap exclusion.

Noisy-gloss retrieval is split-dependent.

A 6-layer text→gloss Transformer (dev BLEU-4 13.17) yields noisy-gloss retrieval sacreBLEU 12.7 on PHX-public (strong-NMT), versus 1.6 for a weak NMT (BLEU-4 1.15) and 6.1 for XLM-R embedding retrieval; donor exclusion reduces the strong-NMT value to 10.9 at τ=0.5, and the same control reaches 0.4 on the template-disjoint holdout (Table S15).

PHX-public pure-donor donor-exclusion τ-sweep.

To test whether the headline reversal survives donor exclusion on the same split, we applied a source-text Jaccard τ-sweep and exact-text-exclusion to the canonical pure text-nearest donor on PHX-public under the original SLRTP2025 BT (Table S12). The reversal survives exact-text exclusion (23.0 vs REC 12.8, 32/641 donors replaced) and source-Jaccard exclusion up to τ=0.5 (17.8 vs REC 12.8); only at τ=0.3 and τ=0.1 does the retrieval-vs-REC gap turn negative.

As a same-paradigm proxy for the SLRTP2025 winner (whose checkpoints are not public), the noisy-gloss retrieval+LM system shows the same pattern: it survives exclusion up to τ=0.5 (10.9, still 6.8× the PT-only baseline and 11.7× the random-donor mean; mean selected-donor Jaccard 0.46) and collapses only at τ=0.1.

10 Template-Density, Holdout Boundary, and Donor-Exclusion Analyses

Template density does not gate the reversal.

Stratifying the 641 test sequences by template density (gloss bigram-Jaccard with nearest train donor), retrieval exceeds REC at all levels: low $+10.7$, mid $+10.1$, high

+12.4 BLEU points (Table S13). Density moderates magnitude but does not gate presence.

Holdout boundary controls.

Truncating to 64/32 frames shrinks the gap from +11.0 to +10.5/+5.6 but the reversal persists. Four holdout controls (Table S14) all fail to reproduce the reversal: the same-evaluator seen-holdout gives REC 77.0 (readout), BT-retrained on unseen holdout gives gap −0.9, retrained BT on PHX-public gives gap −0.4, and cross-fit A/B gives gap −0.1.

Frozen template-holdout with retrained components.

A deterministic template-holdout (14,988 train-core windows, 1,538 holdout windows, disjoint by window, source sequence, and exact gloss template) with all components retrained on train-core shows text-nearest retrieval above random (+2.3) but far below REC (−10.8): the shortcut is real over random but does not exceed REC on the clean split. Donor-exclusion sweeps show the advantage persisting at τ=0.5 and collapsing only at τ=0.1 (Table S15).

The SLRTP2025 test vocabulary is nearly closed (95.4% gloss types in train); nearest donors share mean gloss unigram containment 0.557 and bigram-Jaccard 0.302, but Progressive Transformer outputs show no donor-proximity effect (ρ=−0.06, p=0.12), versus +0.29 for retrieval.

11 Transcript-Slot Diagnostic Details

This appendix documents the rule-based German weather slot extractor referenced in the main text. The extractor identifies four slot families (number/temperature, time, place, weather-event) using a frozen lexicon and normalization rules. On a 50-reference audit (seed 20260729), per-family precision was 0.91–1.00 and recall 0.75–1.00 (overall precision 0.97, recall 0.86); errors are dominated by recall misses from inflections, compounds, and lexicon gaps. A stratified audit of 50 decoded hypotheses (25 REC-decode, 25 PURE-decode, seed 20260730) showed precision nearly equal (0.96 vs 0.97) but recall lower on PURE-decode (0.83 vs 0.91), because donor-transcript-style outputs use more inflected verbs and compounds (*regnet*, *unwetterwarnungen*) that the lexicon misses. The measured PURE slot-F1 is thus attenuated more than REC's, so the PURE>REC contrast is, if anything, underestimated. The audit used a single auditor; inter-annotator agreement is unavailable. The full lexicon, normalization rules, and per-family precision/recall are in the artifact.

12 Competence Extrapolation Details (Exploratory)

This appendix gives the full OLS/GP extrapolation analysis referenced in the main text (RQ1). We emphasize that this is exploratory and the prediction intervals are descriptive visual guides, not calibrated inferential tests, because the 42 unique checkpoints share data, code, and hyperparameters (non-IID) and the target point (dev BLEU-4 = 13.38) lies outside the observed range.

Method.

We fit the gap–competence dose–response over all 42 unique non-released weight binaries (the 43 decoded runs reduce to 42 after SHA-256 deduplication of the long-schedule/validation-freq-misread seed-202 duplicate), spanning dev BLEU-4 0–9.8 (37 checkpoints at ≥ 4.5; the five degenerate checkpoints sit at dev ≈ 0), using three models: OLS linear, OLS quadratic, and a Gaussian Process with RBF + white kernel. Each model predicts the PURE–REC gap at dev BLEU-4 = 13.38.

Results.

Table S16 reports the predictions and 95% PIs at dev BLEU-4 = 13.38. All predictions are far below the observed canonical +10.24. Table S17 reports a leave-one-family-out cross-validation; the OLS quadratic prediction is the most sensitive (shifts from −1.09 on all 42 checkpoints to +2.88 when restricted to the 37 non-degenerate checkpoints with dev ≥ 4.5, and to +0.23 when the 19 distillation-family checkpoints are removed), while GP predictions stay negative with leave-one-family-out upper bounds below +0.3.

Caveats.

First, the GP upper bound (+0.3) partly reflects RBF kernel prior shrinkage toward the mean in the extrapolation region (sparse data beyond dev BLEU-4 9.8), not purely data evidence. Second, the OLS quadratic fit is sensitive to the distillation family and to the inclusion of the degenerate checkpoints. Third, the 42 checkpoints are not IID. We therefore report these fits only as descriptive visualizations and explicitly avoid claiming that "competence alone is unlikely to explain the reversal" on the basis of these fits. The unobserved $(9.8, 13.38)$ interval could in principle harbor non-linear behavior that the present experiments cannot rule out.

Per-item gap decomposition (experiment C).

To characterize the *mechanism* rather than the identity question, we regress the per-item sentence-level PURE–REC gap y_i (sacreBLEU sentence BLEU-4 under the released evaluator, canonical donor registry; mean 8.4, SD 18.7, $n = 641$) on interpretable item-level features: donor–reference token-set Jaccard, normalized character Levenshtein between donor and reference, an exact-match-train-text flag, the number of training texts sharing the reference's 60-character prefix (template density), the number of training texts with Jaccard ≥ 0.5 to the reference (high-overlap-train), reference text length, donor reuse count, a confidence=1 flag, and a 9-level signer one-hot block (Fig. S3). The full model explains $R^2 = 0.33$ of the per-item variance. The strongest single-feature contributions, measured by leave-one-out R^2 drop, are high-overlap-train (0.093), template density (0.051), and donor–reference Levenshtein (0.039); the donor–reference Jaccard contributes 0.007, the exact-match flag 0.001, and the entire signer block 0.008 — signer identity is essentially irrelevant.

The signs of the standardized coefficients are informative. Items where the donor is lexically close to the reference (high Levenshtein similarity, $\beta^* = +6.1$; high donor–reference Jaccard, $\beta^* = +2.9$) show *larger* gaps, whereas items with many high-overlap training neighbors ($\beta^* = -7.6$) or high template density ($\beta^* = -7.3$) show *smaller*

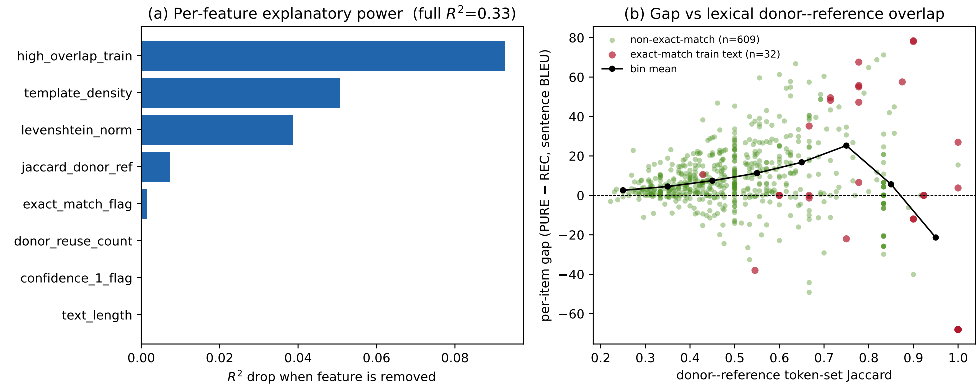

Fig. S3: Per-item gap decomposition (experiment C). (a) Per-feature R^2 drop when the feature is removed from the full OLS model ($R^2 = 0.33$, $n = 641$). High-overlap-train count and template density are the strongest single contributors; the full signer block contributes $\leq 1\%$. (b) Per-item gap vs donor–reference token-set Jaccard; red marks the 32 items whose reference text exactly matches a training text, green the remaining 609; the black line is the bin mean. The gap grows with donor–reference lexical overlap, consistent with the lexical-coupling mechanism. Outcome is sentence-level BLEU-4 (sacreBLEU 2.5.1, 13a, exp smooth); sentence-level mean need not equal corpus BLEU.

gaps. The reversal is therefore largest on items where the retrieval criterion (source-text Jaccard) happens to deliver a donor that matches the reference register closely, but where the recorded poses do not independently let the recognizer recover that text — precisely the lexical-coupling mechanism predicted by the construction (the original reference is a speech transcription of the same source sentence the donor was selected by). Signer identity and text length contribute negligibly; the unexplained 67% of variance is consistent with per-item pose variability and BT-decoder stochasticity that these surface features do not capture. This decomposition does not identify a hidden-pipeline detail; it localizes the reversal to a lexical-coupling property that is constructible on PHX-public's template-dense, high train–test overlap register, consistent with the corpus-level audit of Alkain et al. Alkain et al. (2026).

13 Supplementary CSL-Daily Boundary Control

CSL-Daily (Chinese Sign Language) serves as a non-template-domain control, but its recognizer is too weak for robust claims. Under Mode B (1,176 sequences), text-nearest donor copy reaches only 1.6 sacreBLEU vs REC 26.7 (Table S18). A readout-ratio check (64 train + 64 test windows) gives train≈test BLEU-1 (0.0019 vs 0.0021, ratio ∼0.9), far below PHX's 6.1: the readout signature does not generalize beyond the released PHX evaluator.

14 Leakage Sanity Checks for Train-Pool Readout

This appendix documents the five sanity checks referenced in §4.2 that rule out implementation-level artifacts as explanations for the released evaluator's 78.8 BLEU / 70.7% EM train-pool readout.

Experiment A: Code-path audit.

The `back_translate()` function in `src/models/back_translate.py` (lines 362–426) calls `model.run_batch(translation_beam_size=3, translation_beam_alpha=-1)`, which invokes `bt_search.beam_search()`. The decoder input is seeded with `<bos>` only and extended autoregressively with the model's own previously generated tokens. The reference text is used exclusively in the post-hoc BLEU/EM scoring step; it never enters the model forward pass. We verified this by code inspection and by confirming that the same model loaded in a fresh Python process produces identical outputs.

Experiment B: Pose–text permutation.

We decoded the first 1,000 training items with beam=3 (BLEU 78.61, EM 70.1%), then permuted the reference labels across items to break pose–text correspondence. Three random permutations yield:

Experiment B′: Input-side pose permutation with output-equivariance.

The label permutation above shuffles reference labels *after* decoding, so it cannot exclude ID-based or cache-based retrieval (a cache lookup by ID would also produce low BLEU when labels are shuffled). To properly test pose conditioning, we held item IDs and text metadata fixed, shuffled the *pose tensors* across 100 training items *before decoding*, and performed output-equivariance verification. For all 100 items, the hypothesis produced when pose_j is decoded in slot i's position exactly matches the hypothesis originally produced for pose_j in its own slot j (100% exact string match). Output length follows the pose tensor, not the slot ID (100% donor-length match vs. 5% slot-length match). This rules out a pose-ignorant pure ID/cache lookup under the tested decoding path and confirms sensitivity to the substituted pose tensor. Additional degenerate-pose controls (zero, Gaussian noise, single repeated pose for 200 items) all collapse BLEU to <1, and reversed batch order reproduces the original BLEU exactly, confirming per-item determinism.

Experiment C: Determinism.

Beam search with no stochastic sampling produces deterministic outputs. We record SHA-256 hashes of all per-item decoded hypotheses in the artifact for cross-process verification.

Experiment D: Stratified EM.

Exact-match rate is high and broadly distributed, not concentrated in a single subgroup: Even unique training texts (appearing exactly once) achieve 70.2% EM, confirming the readout is not driven by template repetition alone. Of the 7,060 hypotheses, 74.8% match some training text exactly (70% match exactly one—their correct paired reference).

Experiment E: Same-signer held-out control.

We decoded 500 test-split poses (unseen during training) from the same 9 signers present in the training pool. This is a same-signer held-out control, not a fully matched membership control: the test items are not matched on length, text-similarity, broadcast date, or template density, so the train–test gap reflects a combination of training-pool exposure and distributional shift. The same signers appear in both splits. The 67-point BLEU gap between seen and unseen items is consistent with training-pool exposure contributing to the readout, but without full covariate matching (length, text-similarity, template density), we cannot attribute the gap solely to exposure vs. distributional shift.

15 Pre-Exclusion Materialization Sensitivity

The canonical §3.2 donor registry applies exact-normalized-text exclusion. An earlier frozen materialization predating this exclusion gives PURE 23.79 (gap +11.01) instead of the canonical 23.02 (gap +10.24). We retain it only as a sensitivity analysis.

Per-item provenance.

On 610/641 items (95.2%) both materializations produce identical donor choices. On the 31 differing items, a post-hoc candidate search found that 30 had at least one excluded Jaccard=1.0 candidate whose released-evaluator decode matched the earlier hypothesis, consistent with the exclusion rule explaining the difference on those items; one item remained unresolved (`results/donor_identification_31.json`).

14-reconstruction sensitivity.

Re-decoding all 14 reconstruction checkpoints under both materializations gives mean $|\Delta\text{gap}| = 0.65$ BLEU (max 1.45). The sign of the gap is unchanged on every checkpoint, and the 14-seed prediction interval remains strictly negative under both materializations, so the non-replication conclusion is unaffected. All headline numbers in the main text and in Sup. G use the canonical materialization (SHA-256 `9170a53026ab3263b451a3632a9e02318745acabf12f0b679921477afbefa301`); this appendix is the only place the pre-exclusion values appear.

16 BLEURT-Substitute Validation

Canonical BLEURT-20 ships a TensorFlow checkpoint that does not load cleanly in our PyTorch-only environment. As a non-standard substitute we use `Elron/bleurt-large-512` for one secondary reference-sensitivity check only (the BLEURT row of the reference-frame analysis). It is not used for any primary claim, table, or figure, and is not described as equivalent to BLEURT-20.

Validation.

On 100 German sentence pairs we scored both canonical BLEURT-20 (via the reference TensorFlow runtime in an isolated CPU container) and the substitute. The substitute has Spearman rank correlation $r = 0.87$ against canonical BLEURT-20 and a mean signed bias of -0.47 (substitute scores lower on average). The bias is roughly uniform across score levels and does not change the qualitative reference-sensitivity conclusion (BLEURT is less sensitive to reference replacement than BLEU, consistent with Czehmann et al. Czehmann et al. (2026)). Full per-pair scores, the container hash, and the substitute model revision are in the artifact.

Scope limit.

Because the substitute is not BLEURT-20, the BLEURT row is demoted from the main metric panel to this appendix. Readers should treat the absolute BLEURT-substitute values as approximate; only the directional reference-sensitivity finding is reported.

17 Additional Protocol Details Moved from the Main Text (Sup. Q)

Text-nearest tie-breaking.

Jaccard ties are resolved by minimum character-level Levenshtein distance and then by minimum SHA-256 hash of the training key. The retrieval algorithm also excludes donors whose NFKC-normalized text equals the query's (exact-normalized-text exclusion).

Scorer equivalence.

The vendored official scorer (SacreBLEU 1.4.2) and the paper scorer (SacreBLEU 2.5.1, `tok:13a`, `smooth:exp`) give identical corpus BLEU on all 28 evaluation cells (maximum $|\Delta| = 0.0000$; byte-identical hypotheses, all n-gram orders non-zero). The main text reports the 2.5.1/13a signature as primary.

Metric implementations.

All secondary metrics use the official SLRTP2025 implementations: they reproduce the official repository values (BLEU-4 12.777, chrF 34.585, WER 85.775, ROUGE 35.196 Walsh et al. (2025)) exactly. Implementation choices matter for chrF (corpus-level sacrebleu `CHRF` defaults; a sentence-level variant yields 48.05 instead of 34.59) and WER (jiwer 3.1.0 with official normalization; raw whitespace-split yields 79.26 instead of 85.78). BLEU-4 is invariant; the official implementations are fixed throughout.

Audit design overview (moved from the main text).

The main-text audit design figure (SI Fig. S4) fixes a set of pose outputs and scores them under two interventions: the evaluator checkpoint (released, fourteen recipe reconstructions, heterogeneous diagnostic checkpoints) and the reference frame (speech-transcribed vs. human sign-to-text).

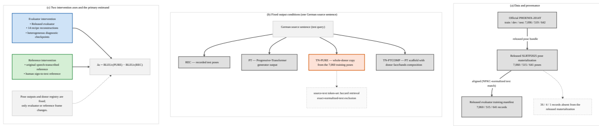

Fig. S4: Audit design: a fixed set of pose outputs is scored under two interventions. (a) Data and provenance: the official PHOENIX-2014T split (7,096/519/642) was materialized by the SLRTP2025 release as 7,060/515/641 poses, matching the released evaluator's training manifest (36/4/1 official records absent). (b) Fixed output conditions for one German source sentence: recorded test poses (REC), Progressive-Transformer output (PT), whole-donor copy TN-PURE (source-text token-set Jaccard, exact-normalized-text exclusion), and PT-scaffold composition TN-PTCOMP. (c) Two intervention axes: the evaluator (released, fourteen recipe reconstructions, heterogeneous diagnostic checkpoints) and the reference frame (speech-transcribed vs. human sign-to-text), with pose outputs and the donor registry fixed; the primary estimand is $\Delta_c = \mathrm{BLEU}_c(\mathrm{PURE}) - \mathrm{BLEU}_c(\mathrm{REC})$.

18 Main-Text Tables (Moved from the Main Text; Sup. P)

The tables in this section were moved from the main text to satisfy the journal page limit; the main text references them as SI Table S22–S33.

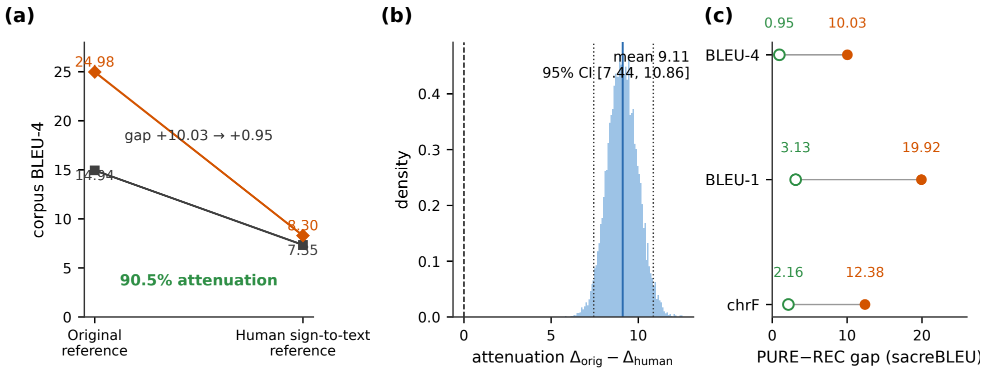

Fig. S5: Human sign-to-text references remove most of the replay advantage on the matched high-confidence subset ($n = 461$; released evaluator, canonical hypotheses). (a) Corpus BLEU-4 under original vs. human references: REC 14.94 → 7.35, PURE 24.98 → 8.30; the gap attenuates +10.03 → +0.95 (90.5%). (b) Bootstrap distribution of the attenuation contrast (10,000 paired resamples; mean 9.08, 95% CI [7.44, 10.86]). (c) PURE−REC gap under original vs. human references across BLEU-4, BLEU-1, and chrF. References are back-translations by a single deaf fluent L2 DGS signer Czehmann et al. (2026); replacement cannot distinguish reference-text defects, paraphrase shift, BLEU floor effects, or disruption of the lexical coupling.

19 Reconstruction Sanity Checks (Reviewer Round-2)

This appendix documents the three sanity checks requested by the reviewer to rule out a systematic implementation defect shared across all 78 training runs. Scripts: `scripts/e_overfit_test.py`, `scripts/e_checkpoint_stats.py`, `scripts/e_inference_regression.py`. Results: `results/overfit_test.json`, `results/checkpoint_stats.json`, `results/inference_regression.json`.

Small-sample overfit test.

A reconstruction model was trained on 50 (and separately 100) randomly sampled training sequences for ≈200 epochs using the identical pipeline as the paper-derived reconstruction runs. On the 50-sample set, teacher-forced NLL fell from 7.66 to 0.009 and free-decode sacreBLEU-4 reached 100.00 under beam-3 decoding. On the 100-sample set, free-decode BLEU also reached 100.00 (300 epochs). The reconstruction code can therefore overfit small data to near-perfect free-decode BLEU, ruling out a fundamental defect in checkpoint loading, tokenizer, padding-mask, EOS, loss normalization, or the beam-search decoding path.

Unified inference regression.

Released and reconstruction checkpoints are loaded through the identical `make_back_translation_model()` function and decoded with the identical `back_translate()` pipeline. All 15 checkpoints (released + 14 reconstructions) share

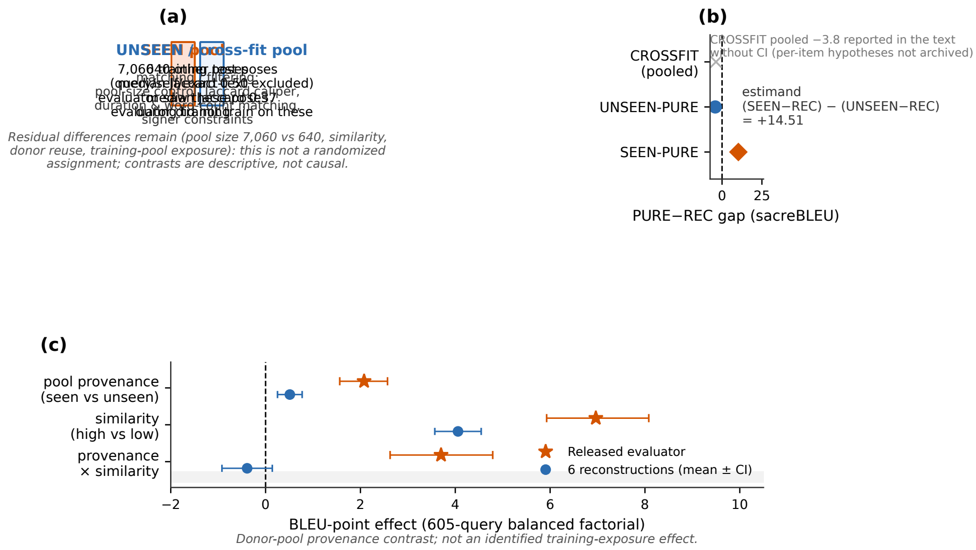

Fig. S6: The replay advantage co-occurs with training-pool donor provenance, but provenance and similarity remain confounded. (a) Donor-construction gaps relative to local REC: SEEN-PURE +10.24; UNSEEN-PURE −4.27 (paired-item CI [−5.39, −3.20]); the pool-substitution estimand (SEEN − REC) − (UNSEEN − REC) = +14.51; CROSSFIT pooled −3.8 (text only, no CI). (b) Balanced 605-query factorial: released evaluator (star) vs. six-reconstruction means (circles) for the pool-provenance and similarity main effects and their interaction. Donor-pool provenance contrast; not an identified training-exposure effect.

byte-identical `txt.vocab` and `gls.vocab` files (SHA-256 confirmed). The released checkpoint decoded through this path reproduces the canonical REC value exactly (12.78 on 641 PHX-public items). The tested reconstruction seeds (101, 202, 303) decode REC at 9.24, 8.22, 9.10 respectively, matching the main-text table. Inference code path is identical for all checkpoints.

Teacher-forced vs. free-decode diagnostics.

Table S36 reports per-checkpoint teacher-forced token accuracy, NLL per token, and free-decode BLEU. Teacher-forced values are on 500 randomly sampled training items; free-decode values are full-pool (7,060 train / 515 dev) under the uniform beam-3 protocol (the per-seed training-time free-decode values in `results/checkpoint_stats.json` were computed on a 500-item subset with a length-50 cap and are not comparable).

Key observation: the released evaluator achieves 96.5% teacher-forced token accuracy on training data versus 55–60% for reconstructions. On the development set, reconstructions achieve slightly lower teacher-forced NLL (2.63–2.68 vs. released 3.24, consistent with less overfitting) but comparable token accuracy (45.8–47.4% vs.

48.6%). The gap in free-decode BLEU is visible already in teacher-forced training-set metrics, not solely a decoding phenomenon.

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Walsh, H., E. Fish, O. Mercanoglu Sincan, M.I. Lakhal, R. Bowden, N. Fox, B. Woll, K. Wu, Z. Li, W. Zhao, H. Wang, W. Zhou, H. Li, S. Tang, J. He, X. Wang, R. Zhang, Y. Wang, L. Cheng, M. Tasyurek, T. Kiziltepe, and H.Y. Keles 2025. SLRTP2025 sign language production challenge: Methodology, results, and future work. In *Proceedings of the Third Sign Language Recognition, Translation and Production Workshop at CVPR 2025*. arXiv:2508.06951.

Table S2: Evidence-graded analysis registry. Rows are grouped by the role each analysis plays in the argument, not by when it was added.

Evidence role	Analysis	Status	Multiplicity handling
Selected probe (descriptive)	PURE–REC gap, original evaluator (post-selection)	D	descriptive MC estimate $p \leq 10^{-4}$; post-selection—see probe multiplicity table in main §4.1
Confirmatory	605-query donor-pool factorial (cells, effects, interaction)	C	Holm across 3 contrasts
	Template-holdout, BT-retrain, cross-fit	C	paired bootstrap CIs
	Protocol checks (scorer, FPS, decoding sweep)	C/D	exact-match / descriptive
	Frozen-protocol confirmation (seeds 1506/1607)	C	confirms family-side pattern
Triangulating	14 reconstruction gaps + prediction interval	C/D	DiD CIs; PI descriptive
(Finding 1: non-replication)	Cross-metric gap table	D	direction/sign only
	Conditioning controls (retrievers, gloss, random)	D	descriptive
	Competence gate, trajectories, Pareto	D	gate counts; 12-threshold grid
	Knowledge-distillation ladder + per-α	D	15 with gap + dev
Triangulating	Reference replacement (single/dual); overlap strata	D	paired bootstrap of gap difference
(Finding 2:	Asymmetric floor-effect control	D	BLEU-4/BLEU-1/chrF drops
reference sensitivity)	Evaluator×reference factorial (7×3)	D	descriptive decomposition
Triangulating	Five diagnostic views (readout-with-generalization)	D	triangulation, not independent
(Finding 3:	In-sample readout diagnosis (train/dev/test)	D	no identity claim beyond train.pt
readout signature)	Leakage sanity checks (permutation, equivariance)	D	exact-match / permutation
Robustness	Donor-pool substitution; canonical B/C/D path	D	query-index bootstrap CIs
	Alternative paths, Shapley, S4 system; balance/caliper/signer	D	path-sensitivity display
	Rescue families (20); big-arch; long-schedule	D	competence attempts
	Released-checkpoint perturbation sweep	D	lockstep collapse with competence
	IPW pool-origin robustness	D	balance robustness check
	Rank stability (9 systems × 7 evaluators)	D	τ_b/ρ/flip means
	Competence extrapolation (OLS/GP)	D	non-IID; LOO CV (5 folds)
	Per-seed extension cells (8 seeds)	D	descriptive; PI only
	Oracle/white-box transfer matrix	D	permutation check
	Template-density strata; donor-exclusion τ-sweeps	D	descriptive
	CSL-Daily readout-ratio check	D	second-dataset boundary
	Transcript-slot diagnostic + extractor audit	D	extractor P/R by system
Administrative	Ethics/licensing declarations	—	updated each revision

Table S3: Canonical checkpoint registry. 78 planned training runs, all 78 completed training. "Beam-3 evaluation" means a run was beam-3 decoded on at least the development set; of these 78 runs, 49 also have a canonical PHX-public PURE–REC gap under the §3.2 donor registry (the remaining 29 were beam-3 evaluated on dev only and lack a canonical PHX-public PURE–REC gap). The 49 decoded runs reduce to 48 unique weight binaries: the long-schedule seed-202 artifact is byte-identical to the validation-freq-misread seed-202 artifact (same wandb run, identical per-step losses, identical best-checkpoint SHA-256) and counts once. 75 non-holdout runs have dev metrics; 52 of these are non-distillation and enter the competence gate. 48 unique runs have both dev BLEU-4 and PHX-public PURE–REC gap (extrapolation-eligible). Plus the released original. Each family row's columns sum to the Total row.

Family	Planned	Trained	Has gap (beam-3)	Has dev	Enters
Reconstructions (primary)	6	6	6	6	multiseed, bootstrap, dose–resp.
Reconstructions (extension)	8	8	8	8	prediction interval, dose–resp.
Rescue lr (A/B $\times$ 4)	8	8	0	8	gate counts only
Rescue expanded	12	12	1	12	gate; wd0 also dose–resp., perturb.
Train-pool ladder	4	4	4	4	dose–resp., gap ranges
Validation-freq-misread	4	4	4	4	dose–resp.
Step-corrected	10	10	10	10	validation-frequency; dose–resp. (2/10)
Large-arch (A1–A4)	4	4	0	4	competence attempt
Confirmation	2	2	2	2	dose–resp., family-side confirm.
Long-schedule	2	2	1	2	gate; best seed dose–resp. (dev 10.02)
BT-retrained	1	1	0	0	holdout series
Cross-fit A/B	2	2	0	0	holdout series
Distillation students	15	15	13	15	ladder, dose–resp.
Total	78	78	49	75	
Released original	—	—	1	1	audit target (not trained)

Per-family columns sum to the Total row (e.g., Has gap: 6+8+4+10+2+1+0+13+4+1+0+0+0 = 49). Gate-eligible = 52 non-distillation runs with dev metrics. Extrapolation-eligible = 48 unique runs with both dev BLEU-4 and PHX-public PURE–REC gap (49 decoded runs minus the long-schedule/validation-freq-misread seed-202 duplicate); the dose–response figure caption uses the subset with dose-response data (14 reconstructions + 4 ladder + 4 validation-freq-misread + 2 step-corrected + 2 confirmation + 1 long-schedule + 15 distillation + 1 wd0 rescue; 2 of the 10 step-corrected seeds have dose-response data). All 15 distillation students completed training and beam-3 evaluation. The 1 wd0 rescue with PHX-public gap is the best rescue checkpoint (weight-decay 0, seed 202); the other 11 rescue-expanded runs have dev metrics but were not decoded on PHX-public. The second long-schedule seed was trained but not decoded on PHX-public. The registry counts all 78 training runs; the artifact ships 51 completed `best.ckpt` files from this registry plus the released checkpoint, while 24 runs' weights are withheld for size (rescue-lr 8, rescue-expanded 11, large-arch 4, long-schedule second seed 1)—their dev metrics and decoded cells remain in the registry and `results/`—and the holdout-series runs (BT-retrained, cross-fit) produce no checkpoints.

Table S4: Spearman correlations across the 31 checkpoints with uniform full-pool readouts (30 constructible plus the released evaluator). n is the number of checkpoints entering each correlation; "non-degenerate" excludes the three α=1.0 distillation students that decode to empty output.

Pair	ρ	p	n
Readout vs. dev BLEU-4	0.916	5.0×10^{-13}	31
Readout vs. dev BLEU-4 (non-degenerate)	0.873	2.9×10^{-9}	27
Readout vs. train−dev BLEU-4	0.040	0.83	31
Readout vs. PURE−REC gap	−0.082	0.66	31
Gap vs. dev BLEU-4	−0.120	0.52	31
Readout vs. train NLL	−0.561	0.030	15
Readout vs. train token accuracy	0.509	0.052	15
Train−dev BLEU-4 vs. gap	−0.053	0.78	31

Table S5: Path sensitivity of the donor-pool substitution decomposition under the original evaluator (common 622-query support; 10,000-resample query-index bootstrap CIs). S1=SEEN-PURE (7,060-pool, max-Jaccard), S2=SEEN-RAND640 (640-pool, max), S3=SEEN-MATCHED (7,060-pool, Jaccard-matched), S4=SEEN-RAND640-MATCHED (640-pool, matched), U=UNSEEN-PURE (test pool). Every ordering telescopes to the same total (A) = S1−U = +14.7.

Ordering	Step	Size contrast	Selection contrast	Origin contrast
Canonical (prespecified)	S1→S2→S3→U	+7.2 [+5.9, +8.4]	−0.8 [−1.8, +0.2]	+8.4 [+7.3, +9.4]
Alternative 1	S1→S2→S4→U	+7.2 [+5.9, +8.4]	+1.8 [+1.1, +2.5]	+5.8 [+4.7, +6.8]
Alternative 2	S1→S3→S4→U	+2.6 [+1.7, +3.5]	+6.4 [+5.1, +7.7]	+5.8 [+4.7, +6.8]
Shapley average	over orderings	+4.9 [+4.0, +5.8]	+4.1 [+3.3, +4.9]	+5.8 [+4.7, +6.8]

Table S6: Factorial design and balance (605 targets). SMD = standardized mean difference, seen vs unseen within stratum.

Quantity	seen_high	seen_low	unseen_high	unseen_low
LaBSE similarity (mean)	0.813	0.654	0.776	0.622
Duration (frames, mean)	100.2	99.5	99.2	98.8
Word count (mean)	13.3	13.6	13.3	13.4
Same-signer rate	100%	100%	100%	100%
Similarity SMD (seen − unseen)	+0.41 (high)			+0.38 (low)

Table S7: Full 2×2 exposure factorial (605 targets; corpus sacreBLEU in BLEU points; NLL per token). CIs: 10,000-resample bootstrap (original) and seed-level t across six reconstructions; Holm correction across three contrasts.

Metric	Evaluator	seen_hi	seen_lo	unseen_hi	unseen_lo	seen main	sim. main	inter.
BLEU	Original	11.12	2.31	7.19	2.08	**+2.08** [+1.6, +2.6]	+6.96 [+5.9, +8.1]	**+3.70** [+2.6, +4.8]
BLEU	Seeds (mean)					+0.51	+4.06	−0.39
NLL	Original	3.74	4.30	3.93	4.37	−0.128	−0.498	−0.124
NLL	Seeds (mean)					−0.064	−0.179	−0.004

Table S8: Multivariate donor balance (622-query common support). All $|SMD| < 0.12$.

Covariate	SEEN-MATCHED mean	UNSEEN mean	SMD	Per-query overlap
Source-text Jaccard	0.411	0.415	−0.021	$\lvert\Delta J\rvert \leq 0.05$: 96%
LaBSE cosine	0.782	0.774	+0.068	—
Duration (s)	3.76	3.74	+0.014	$\lvert\Delta\rvert \leq 1$ s: 57%
Word count	12.29	12.31	−0.003	$\lvert\Delta\rvert \leq 2$: 69%
Template density	0.310	0.326	−0.075	—
Same signer as query	0.183	0.228	−0.111	both same-signer: 4.7%

Table S9: Epoch-validation approximation (validate every 14 epochs, not steps): 300 epochs, dev-NLL selection. All four seeds produce negative PURE–REC gaps under the canonical §3.2 donor registry; none shows the reversal or readout signature.

Seed	ep.	dev NLL	REC	PURE gap
101	27	2.11	10.12	−0.60
202	—	—	10.14	−1.70
303	—	—	8.74	−1.51
404	—	—	8.43	−0.52
Released evaluator	1,820	—	12.78	+10.24

Table S10: Pose-perturbation battery (corpus sacreBLEU, BLEU points) under the released evaluator and the best rescue checkpoint (wd0, seed 202). The TN-PURE-v1 identity column here (23.79) reflects the pre-exclusion materialization used at the time this battery was decoded; the canonical materialization gives 23.02 (gap +10.24). Because the battery varies the pose and re-decodes, not the donor choice, the perturbation differentials are unaffected by the materialization difference (Sup. N: mean $|\Delta\text{gap}| = 0.65$, max 1.45).

Evaluator	System	identity	reversal	window shuffle	jitter 5%	hand mask	face mask
Original	REC-v1	12.78	7.25	11.97	12.51	1.35	3.15
Original	TN-PURE-v1	23.79	7.72	18.35	23.91	1.30	4.06
wd0 (seed 202)	REC-v1	10.73	6.26	10.54	10.60	0.63	1.58
wd0 (seed 202)	TN-PURE-v1	10.98	6.66	9.65	10.92	0.58	1.43

Table S11: Cluster-aware bootstrap for the headline PURE–REC gap (canonical §3.2 registry, original refs, released-evaluator hypotheses from `results/gap_43_canonical_beam3_items/released_{gt,pure}.json`). All CIs are 10,000-resample sacreBLEU `tok:13a` exp-smooth over precomputed per-item n-gram sufficient statistics. With so few clusters (especially show with only 2), these are *exploratory sensitivity checks*, not confirmatory tests. Donor reuse: 641 queries select 565 unique donors (61 donors serve more than one query; maximum reuse 6; Gini 0.11).

Analysis	Design	Clusters	Gap	95% CI
Query-level (baseline)	item	641	+10.24	[+8.88, +11.62]
Donor cluster	donor	565	+10.25	[+8.87, +11.64]
Donor two-way (query×donor)	nested	565	+10.25	[+8.79, +11.74]
Signer	signer	9	+10.30	[+9.29, +11.39]
Broadcast show	show	2	+9.27	[+5.16, +11.44]
Broadcast date	date	297	+10.25	[+8.83, +11.72]
Signer×Show	combined	17	+10.14	[+7.89, +11.91]

Table S12: PHX-public pure-donor source-text donor-exclusion τ-sweep under the original SLRTP2025 BT (canonical tie-breaking, 12.5 fps; sacreBLEU on the 0–100 scale). The retrieval-exceeds-REC reversal survives exact-text exclusion and τ=0.5 exclusion.

Condition	sacreBLEU	Δ vs REC	Reversal?
REC baseline	12.8	—	—
Text-nearest pure (no exclusion)	23.8	+11.0	YES
Text-nearest pure (canonical, w/ exclusion)	23.02	+10.24	YES
Exact-text excluded	23.0	+10.2	YES
τ=0.7 (Jaccard > 0.7 excluded)	21.2	+8.4	YES
τ=0.5 (Jaccard > 0.5 excluded)	17.8	+5.0	YES
τ=0.3 (Jaccard > 0.3 excluded)	8.0	−4.8	NO
τ=0.1 (Jaccard > 0.1 excluded)	0.2	−12.5	NO
Random donor (mean of 20 seeds)	0.9	−11.6	NO

Table S13: Template-density stratification (sacreBLEU 0–100). Retrieval exceeds REC at all density levels.

Density stratum	Mean density	REC	Retrieval	Gap
Low (n=249)	0.152	5.9	16.6	+10.7
Mid (n=179)	0.254	9.5	19.7	+10.1
High (n=213)	0.510	26.1	38.6	+12.4

Table S14: Holdout boundary controls (sacreBLEU, 0–100). None reproduces the +11.0 reversal.

Control	Setup	REC	Retrieval	Gap
Same evaluator	test split (template-overlap)	12.8	23.02	+10.24
Same evaluator	template-holdout (seen in training)	77.0	20.1	−56.9
BT-retrained (unseen holdout)	train-core 6,535	6.2	5.4	−0.9
Retrained BT on PHX-public	unseen to both	8.7	8.3	−0.4
Cross-fit A/B (donor-unseen)	pooled, n=659	4.0	3.9	−0.1

Table S15: Frozen template-holdout controls (1,538 windows, α=0.5; sacreBLEU 0–100).

System	sacreBLEU	Empty %	Mean Jac.
REC baseline	13.10	9.95	—
Random train donor	0.04	11.83	—
Text-nearest train donor	2.30	10.66	—
LaBSE / multi-SBERT (donor copy)	1.73 / 1.51	9.2 / 11.3	—
Noisy-gloss retrieval (strong NMT)	0.40	10.90	—
Retrieval composition (α=0.5)	2.20	10.79	—
τ=0.1 / 0.3 / 0.5 / 0.7 exclusion	0.04 / 1.05 / 1.93 / 2.01	10.0–10.5	0.10/0.29/0.40/0.42
BM25 / TF-IDF nearest	1.75 / 2.36	10.8 / 8.8	—

Table S16: Competence extrapolation: gap predictions at dev BLEU-4 = 13.38 from three fits over 42 unique non-released binaries (dev BLEU-4 0–9.8; 37 at $\geq$ 4.5). All predictions are far below the observed canonical +10.24. PIs are *descriptive* (non-IID checkpoints); they should not be read as calibrated uncertainty intervals. Sensitivity row: the quadratic restricted to the 37 non-degenerate checkpoints (dev $\geq$ 4.5) shifts to +2.88.

Fit	R^2 / LML	Prediction at 13.38	95% PI (descriptive)
OLS linear	0.48	−1.84	[−2.67, −1.01]
OLS quadratic	0.50	−1.09	[−1.83, −0.35]
Gaussian Process (RBF + white)	−48.1	−1.00	[−2.16, +0.17]
Released evaluator (observed, canonical)	—	+10.24	[+8.88, +11.62]

Table S17: Leave-one-family-out cross-validation of the competence extrapolation. Each row refits all thre models after removing one checkpoint family (ladder fractions are grouped with the distillation family, matchin the §4.2 figure family mapping). The GP upper bound stays below +0.3 across all folds; OLS quadratic is th most sensitive (shifts sign when distillation is removed). All intervals descriptive.

Family removed	n remaining	OLS-lin pred	OLS-quad pred	GP pred	GP upper	OLS-lin R^2
None (full)	42	−1.84	−1.09	−1.00	+0.17	0.48
Reconstructions (14)	28	−1.84	−2.07	−0.89	+0.25	0.57
Distillation (19)	23	−0.78	+0.23	−1.18	−0.41	0.03
Validation-freq-misread (8)	34	−1.89	−1.32	−0.97	+0.22	0.56
Rescue (1)	41	−1.82	−0.97	−0.98	+0.18	0.47

Table S18: CSL-Daily Mode B donor-exclusion (1,176 sequences; 0–100 scale).

Condition	sacreBLEU	BLEU-1
REC baseline	26.7	79.1
Text-nearest donor	1.6	43.5
Random donor	0.0	3.9
τ=0.5 exclusion	1.0	42.6
BM25 retrieval	1.5	39.2
TF-IDF retrieval	1.2	38.5

Table S19: Label permutation test (1,000 items, reference labels shuffled *after* decoding). BLEU collapses from 78.6 to ~1.0, confirming output–reference alignment. This test alone cannot distinguish pose-conditioned output from ID/cache-based retrieval; the input-side pose permutation below addresses that.

Condition	BLEU	EM
Original (correct pairing)	78.61	0.701
Permutation 0	1.06	0.000
Permutation 1	1.30	0.000
Permutation 2	0.90	0.002

Table S20: Input-side pose permutation with output-equivariance (100 items). Pose tensors are shuffled *before* decoding while IDs stay fixed. The equivariance column verifies $E(\text{pose}_j, \text{slot}_i) = E(\text{pose}_j, \text{slot}_j)$.

Condition	BLEU vs slot ref	Detail
Original decode vs own reference	75.62	baseline
Shuffled poses vs slot reference	0.95	BLEU collapses
Shuffled poses vs donor reference	75.62	identical to original
Output-equivariance match rate	—	100/100 (100%)
Donor-length match rate	—	100/100 (100%)
Slot-length match rate	—	5/100 (5%)

Table S21: Stratified exact-match rate on the full 7,060-item training pool under beam=3 free-decode.

Stratum	n	EM
By signer		
Signer01	1,852	0.683
Signer02	83	0.590
Signer03	579	0.736
Signer04	1,049	0.748
Signer05	1,623	0.701
Signer06	35	0.714
Signer07	762	0.696
Signer08	844	0.715
Signer09	233	0.734
By reference length		
Short (≤5 words)	203	0.837
Medium (6–10)	1,319	0.685
Long (11–20)	4,335	0.714
Very long (>20)	1,203	0.686
By text repetition in train		
Unique (1 occurrence)	6,777	0.702
Low (2–3)	57	0.737
Mid (4–10)	76	0.684
High (>10)	150	0.940

Table S22: Same-signer held-out control: train (seen) vs. test (unseen) free-decode. The 67-point BLEU gap is consistent with training-pool exposure contributing to the readout, though the lack of full covariate matching means we cannot quantify how much of the gap is exposure vs. distributional shift.

Split	n	BLEU	EM	Shared signers
Train (seen)	7,060	78.82	0.707	9
Test (unseen)	500	12.19	0.032	9 (same)

Table S23: Decoded-text metric sensitivity under the original SLRTP2025 evaluator (641 sequences). The reversal direction is consistent across decoded-text metrics; teacher-forced NLL does not reproduce it (SI Sup. O). Item-level pose-DTWp is not reported (see main text).

System	BLEU-4	BLEU-1	chrF	ROUGE-L	WER	BERTSc.
REC-v1 (recorded test poses)	12.78	34.40	34.59	35.20	85.77	0.757
PT-v1	1.59	16.91	19.03	17.26	101.68	0.669
TN-PTCOMP	18.86	46.35	41.88	45.17	77.36	0.793
TN-PURE	**23.02**	**55.43**	**47.82**	**53.07**	**71.87**	**0.822**
Gap PURE−REC	+10.24	+20.94	+12.93	+17.87	−13.90	+0.066

Table S24: Asymmetric floor-effect control (full-641 sensitivity analysis). PURE loses a larger fraction of its score REC across all metrics when references are replaced, so simple symmetric compression is insufficient; the asymm does not discriminate between reference-text defects and training-distribution alignment.

Metric	REC (orig)	REC (human)	REC drop	PURE (orig)	PURE (human)	PURE d
BLEU-4	12.78	5.66	7.11 (55.7%)	23.02	6.52	16.50 (71.
BLEU-1	34.40	29.28	5.12 (14.9%)	55.43	32.35	23.08 (41.
chrF	34.59	32.75	1.84 (5.3%)	47.82	35.02	12.80 (26.
Dual-ref gap (orig+human)	BLEU-4: +11.50; BLEU-1: +20.03					

Table S25: PURE–REC gap across metrics and evaluators on PHX-public (641 sequences; canonical decoding) for the original evaluator and the *six primary* reconstructions (of fourteen; the eight extension seeds show the same near-zero/negative pattern in the main-text multiseed factorial table). chrF, WER and ROUGE-L use the official SLRTP2025 implementations. WER and NLL are sign-flipped so positive always means PURE is better than REC. BERTScore-F1 uses `bert-base-multilingual-cased` (bert-score 0.3.13 package, whose module `__version__` string reads 0.3.12; `lang=de`, `idf=False`, no baseline rescaling; model revision logged in the artifact); per-checkpoint BERTScore cells are in the artifact. The large reversal is consistent in direction across decoded-text metrics under the original evaluator and disappears under all six primary reconstructions; residual reconstruction signs are mixed and near zero for n-gram metrics.

Evaluator	ΔBLEU-4	ΔBLEU-1	ΔchrF	ΔWER†	ΔNLL†	ΔBERTSc.
Original	+10.24	+20.94	+12.93	+13.90	−0.419	+0.066
Seed 101	−0.80	−1.23	−1.44	−2.95	−0.219	−0.005
Seed 202	−1.30	−1.33	−1.57	−3.52	−0.233	−0.006
Seed 303	−1.85	−1.99	−2.26	−1.53	−0.240	+0.001
Seed 404	−1.81	−1.60	−1.90	−1.87	−0.194	−0.001
Seed 505	−1.28	−1.53	−1.59	−3.41	−0.268	−0.004
Seed 606	−1.24	−1.26	−1.55	−3.46	−0.169	−0.001

†WER and NLL are sign-flipped: positive means PURE is better than REC (lower WER, lower NLL).

Table S26: Public SLRTP2025 challenge leaderboard on ph14t-test (Walsh et al. Walsh et al. (2025), Table 1), computed under the same released BT evaluator the main text audits; the "Ground Truth" row matches the canonical REC exactly. USTC-MoE is roughly even with REC on lexical BT metrics (BLEU-4 −0.72, BLEU-1 +0.45, chrF +2.24, WER +7.72) despite a much larger DTW-MJE (0.0448) and duration ratio (1.438). This single-evaluator co-occurrence is descriptive and does not establish evaluator retrieval-favoring bias.

System	BLEU-1	BLEU-4	chrF	ROUGE	WER↓	DTW-MJE↓
Recorded test poses (REC)	34.40	12.78	34.59	35.20	85.77	0.0000
Team 1 – USTC-MoE (retrieval)	**34.85**	**12.06**	**36.83**	**36.59**	93.49	0.0448
Team 3 – Hacettepe	30.44	9.59	29.70	30.64	88.88	0.0423
Progressive Transformer	22.17	5.43	24.13	21.98	101.45	0.0418
Team 2 – hfut-lmc	16.96	2.05	25.88	19.77	147.85	0.0403
TN-PURE-v1 (replay probe)	55.43	23.02	47.82	53.07	71.87	0.00678

Table S27: Per-α distillation ladder summary. 15/15 students evaluated with the canonical donor registry; "dev BLEU-4 (max)" is the training-time dev BLEU-4 recorded by the original training protocol (legacy validation-selected protocol; not comparable to the uniform full-pool free-decode dev BLEU-4 reported in main §4.2; checkpoints were selected by training-time dev NLL, Sup. E). α=0 = hard-label only; α=1 = full soft-label KL. All α<1.0 yield negative gaps; α=1.0 produces empty/degenerate outputs (641/641 or 483/641 empty hypotheses). Train-pool readout uses the identical full-7,060, beam-3, [:: 2]-subsample protocol for every student and for the released evaluator (no 1,000-item-prefix mixing); the three α=1.0 students decode to empty or near-empty strings on both dev and test (uniform full-pool: 515/515 or 390/515 dev hypotheses empty, 641/641 or 483/641 test hypotheses empty) and score 0.00/0.0%; their dev BLEU-4 is degenerate and not reported.

α	n	dev BLEU-4 (max)	gap (range)	train readout (BLEU / EM%)	reaches 10.5?
0.0 (hard-label)	3	10.51	$[-1.71, -0.80]$	8.45–8.87 / 1.9%	1/3
0.25	3	10.75	$[-2.01, -0.77]$	7.81–8.63 / 1.5–1.9%	2/3
0.5	3	10.41	$[-1.38, -0.77]$	7.73–8.20 / 1.7–1.9%	0/3
0.75	3	10.93	$[-1.21, -0.81]$	6.59–7.01 / 1.3–1.4%	2/3
1.0 (full KL)	3	*degenerate*	$[0.00, 0.00]$	0.00 / 0.0% (empty)	—
Released (full 7,060, beam=3)	1	13.38	+10.24	78.82 / 70.7%	—

Table S28: Canonical checkpoint registry (summary). 78 training runs across 13 families; 44 non-degenerate checkpoints enter the non-replication count, 49 have a canonical PHX-public PURE–REC gap, and the released evaluator is the audit target. Full per-checkpoint details in Sup. D.

Family	Planned	Trained	Has gap	Has dev	Enters
Reconstructions (14)	14	14	14	14	multiseed, bootstrap, PI, dose–resp.
Rescue (20)	20	20	1	20	gate; wd0 dose–resp.
Other families (21)	21	21	14	20	dose–resp.
Distillation (15)	15	15	15	15	ladder, dose–resp.
Total	70	70	43	67	
Released original	—	—	1	1	audit target

Table S29: Experimental protocol summary (moved from the main text).

Setting	Recognizer	Layout	Split size	N eval	α
PHX-public	SLRTP2025 evaluator	178-joint	641 seq	641	0.88 (tuned)
PHX-holdout-MSKA	mska_slr (phoenix-2014t)	133-joint	1,538 win	1,538	0.50 (frozen)
CSL-Daily	mska (csl-daily)	133-joint	256 of 1,176 win	256	0.50 (frozen)

Table S30: Competence rescue sweep: 20 runs across six hyperparameter families all fail the competence gate. Best variant: weight-decay 0, seed 202 (gate threshold defined in the main text).

Variant family	dev BLEU-4 range	min $\lvert\Delta$BLEU-4$\rvert$ vs Original
lr 5×10^{-4} / 2×10^{-3} (8 runs)	7.9–9.7	3.7
batch size 128 / 512	6.8–9.2	4.2
dropout 0.2	6.9–7.5	5.9
weight decay 0	**9.8–9.9**	**3.5**
label smoothing 0.1	6.8–7.4	6.0
long schedule (600 epochs, lr 5×10^{-4})	8.2–8.3	5.1

Table S31: Donor-origin contrast estimands (sacreBLEU, BLEU points; gaps vs local REC). SEEN = 7,060 training poses; UNSEEN = 640 other test poses. Per-seed cells in the artifact.

Evaluator	SEEN-PURE	UNSEEN-PURE	SEEN−REC	UNSEEN−REC	Estimand
Original	23.02	8.51	+10.24	-4.27	+**14.51**
Seeds (range)	6.83–9.09	6.59–7.98	−1.85 to −0.80	−1.94 to −0.27	−0.15 to +1.22

Table S32: Matched-subset reference sensitivity (corpus sacreBLEU, BLEU points) under the released evaluator. Primary analysis: confidence=1 subset (461 items) with original and human references computed on the *same* items. Sensitivity analysis: full 641 items. Human references: Czehmann et al. Czehmann et al. (2026) back-translations (CC BY-NC-SA 4.0). Confidence breakdown: 461 confidence=1.0, 121 confidence=0.5, 59 confidence=0.0.

Reference condition	N	REC	PURE	Gap
Primary: matched confidence=1 subset				
Original (speech-transcribed)	461	14.94	24.98	+10.03
Human back-translation (conf=1)	461	7.35	8.30	+0.95
Paired attenuation	9.08 BLEU pts (90.5%), CI [7.44, 10.86]; ratio CI [0.81, 1.01]			
Sensitivity: full 641 items				
Original (speech-transcribed)	641	12.78	23.02	+10.24 [+8.88, +11.62]
Human back-translation (full)	641	5.66	6.52	+0.86 [+0.38, +1.91]
Paired attenuation				9.38 BLEU pts (91.6%)
Human refs, reconstructions (6 seeds)	641	3.49–4.62	3.35–4.35	−0.15 to −0.65

Table S33: Cross-evaluator re-decoding of the noisy-gloss retrieval outputs (641 PHX-public keys; §4.5). Gap = noisy-gloss − REC, corpus sacreBLEU-4 (flat-reference form, §3 protocol). The realistic retrieval pipeline is roughly even with REC under the released evaluator (−0.13), and every non-degenerate evaluator gives a negative gap; the five degenerate checkpoints (three α=1.0 distillation students, two smallest ladder fractions) decode empty or near-empty hypotheses.

Evaluator	n	Gap range	Mean gap
Released	1	−0.13 (NG 12.65, REC 12.78)	−0.13
Reconstructions	14	[−1.58, −0.37]	−1.14
Validation-freq-misread / step-corrected / confirmation / long-schedule	9	[−1.84, −0.67]	−1.33
Distillation students (α<1.0)	12	[−1.94, −0.60]	−1.23
Rescue wd0	1	−1.49	−1.49
Ladder (non-degenerate fractions)	2	[−0.58, −0.07]	−0.33

Table S34: Cross-evaluator re-decoding of the rebuilt USTC-MoE fixed outputs (497 of 641 PHX-public keys; §4
Gap = USTC-MoE − REC, corpus sacreBLEU-4. The rebuild decodes far below the officially reported leaderbo
score (2.42 vs. 12.06), so it is pipeline-faithful but not score-faithful to the archived submission; every non-degener
evaluator gives a negative gap. The five degenerate checkpoints (three α=1.0 distillation students, two smallest lad
fractions) decode empty or near-empty hypotheses.

Evaluator	n	Gap range	Mean g
Released	1	−9.94 (USTC-MoE 2.42, REC 12.36)	−9
Reconstructions	14	[−8.30, −5.91]	−7
Validation-freq-misread / step-corrected / confirmation / long-schedule	9	[−8.16, −6.58]	−7
Distillation students (α<1.0)	12	[−8.26, −6.13]	−7
Rescue wd0	1	−7.58	−7
Ladder (non-degenerate fractions)	2	[−5.47, −4.28]	−4

Table S35: Four-layer reproduction of the headline results from the artifact (§3). Layer 1 regenerates every paper number from committed per-item decodes; Layer 2 re-executes analyses from per-checkpoint data; Layer 3 re-decodes fixed outputs from checkpoint weights; Layer 4 retrains from scratch. `make core-audit` executes L1 only and is not an end-to-end reproduction from training data.

Layer	Command	Scope	Does NOT do
L1: number recomputation	`make core-audit`	Regenerates canonical donor registry and all headline numbers from committed per-item decodes (asserts REC 12.78 / PURE 23.02 / gap +10.24 / CI $[8.88, 11.62]$)	Does not re-decode or retrain
L2: analysis re-execution	`make round2-all`; individual `make` targets (donor-bootstrap, ref-frame-figure, experiment-C/A1/D, full-readout)	Re-runs every analysis from committed per-checkpoint data and hypotheses	Does not re-decode
L3: model re-evaluation	`scripts/decode_cells.py`; `scripts/e_full_readout.py`	Re-decodes fixed outputs from checkpoint weights	Does not retrain; requires checkpoint weights + data bundle
L4: from-scratch retraining	`scripts/train_reconstruction.py`; `launch_reconstructions.sh`; rescue/distillation/validation-freq-misread launchers	Retrains and re-evaluates from the public recipe	Requires PHOENIX-2014T licence, exact env, ∼200 GPU-hours

Table S36: Per-checkpoint teacher-forced and free-decode diagnostic statistics. Teacher-forced token accuracy is the fraction of non-pad tokens where arg max of the teacher-forced logits matches the target; free-decode BLEU uses beam-3 autoregressive decoding on the full training pool (7,060) and full dev set (515) under the uniform protocol. Full data in `results/checkpoint_stats.json`.

Checkpoint	Train tok. acc.	Train NLL/tok	Train free-decode BLEU	Dev NLL/tok	Dev free-decode BLEU
Released	0.965	0.158	78.82	3.236	13.38
Seed 101	0.570	1.822	8.54	2.648	8.51
Seed 202	0.550	1.887	7.62	2.675	6.88
Seed 303	0.596	1.723	8.24	2.672	7.72
Seed 404	0.550	1.922	8.06	2.645	7.33
Seed 505	0.564	1.870	8.58	2.655	9.04
Seed 606	0.574	1.816	7.20	2.672	6.49
Seed 707	0.590	1.718	9.25	2.644	9.37
Seed 808	0.574	1.806	9.11	2.654	8.97
Seed 909	0.565	1.851	8.58	2.654	7.84
Seed 1001	0.584	1.767	8.63	2.682	9.33
Seed 1102	0.584	1.759	8.94	2.632	8.67
Seed 1203	0.574	1.793	8.38	2.659	7.45
Seed 1304	0.564	1.852	8.47	2.642	8.93
Seed 1405	0.564	1.826	7.34	2.677	7.22
Reco. mean	0.572	1.815	8.35	2.659	8.13
Reco. range	0.550–0.596	1.718–1.922	7.20–9.25	2.632–2.682	6.49–9.37